

Charcot-Marie-Tooth Disease Axonal Type 2P (CMT2P) — Comprehensive Disease Characteristics Report

Disease: Charcot-Marie-Tooth Disease, Axonal, Type 2P **OMIM:** #614436 | **Gene:** *LRSAM1* (chr9q33.3) | **Category:** Mendelian **Suggested MONDO:** MONDO:0013749 (Charcot-Marie-Tooth disease, axonal, type 2P)

Evidence base: literature-derived (human clinical, in vitro/biochemical, model organism, and computational). No primary datasets were provided; all claims are supported by primary literature (PMIDs) and standard disease/gene databases.

Summary

Charcot-Marie-Tooth disease axonal type 2P (**CMT2P**; OMIM #614436) is a rare, genetically defined form of hereditary axonal sensorimotor peripheral neuropathy caused by pathogenic variants in ***LRSAM1***, a gene on chromosome 9q33.3 encoding a **RING-type E3 ubiquitin-protein ligase**. It is a member of the large, clinically and genetically heterogeneous Charcot-Marie-Tooth (CMT) family — the most common inherited neuromuscular disorder, affecting roughly 1 in 2,500 people worldwide — but CMT2P itself is one of the rarer axonal (type 2) subtypes. The disease is distinguished by a **late/adult onset** (typically second to fifth decade), **slow progression**, **lower-limb predominance**, frequent **neuropathic pain**, and marked **phenotypic variability** that includes asymptomatic mutation carriers.

The molecular pathology centers on **loss of LRSAM1 ubiquitin ligase activity**. Pathogenic variants cluster in the C-terminal RING domain that mediates the essential E2–E3 interaction required for transferring ubiquitin to substrate proteins (the best-validated target being TSG101, a component of the ESCRT endosomal-trafficking machinery). CMT2P displays a distinctive **dual inheritance pattern**: recessive alleles that trigger complete loss of protein (e.g., via nonsense-mediated decay, NMD) versus dominant alleles that escape NMD and produce a truncated protein exerting a **dominant-negative** effect on the RING domain.

Whether a 3'-located premature-termination variant escapes or triggers NMD effectively determines whether disease is dominant or recessive — a rare and instructive example of a single gene where variant position dictates inheritance mode.

There is **no curative therapy**; management is entirely symptomatic and supportive (ankle-foot orthoses, physical/occupational therapy, foot-deformity surgery, neuropathic pain control, and genetic counseling), and **life expectancy is normal**. An emerging and biologically intriguing finding is that LRSAM1 disease may extend beyond the peripheral nervous system: within the original dominant kindred, several affected members developed **late-onset parkinsonism**, and additional cases pair CMT2P with parkinsonian features — linking this peripheral neuropathy to a central nervous system phenotype and to ubiquitin-proteostasis themes shared with neurodegeneration.

1. Disease Information

Overview. CMT2P is an axonal (as opposed to demyelinating) hereditary motor and sensory neuropathy. Clinically it presents as a length-dependent, distal, symmetric sensorimotor polyneuropathy with distal weakness and atrophy (legs > arms), distal sensory loss, reduced/absent tendon reflexes, foot deformity (pes cavus), and gait impairment — the classic CMT phenotype — but with a characteristically **mild, late-onset, slowly progressive** course.

Key identifiers:

Resource	Identifier
OMIM (phenotype)	#614436 (Charcot-Marie-Tooth disease, axonal, type 2P)
OMIM (gene)	<i>LRSAM1</i> 610933
Gene / HGNC	<i>LRSAM1</i> , HGNC:25135
NCBI Gene	90678
UniProt	Q6UWE0
Suggested MONDO	MONDO:0013749
ICD-10	G60.0 (Hereditary motor and sensory neuropathy)
ICD-11	8C20 (Hereditary motor and sensory neuropathy)
MeSH	Charcot-Marie-Tooth Disease (D002607)

Synonyms / alternative names: CMT2P; Charcot-Marie-Tooth disease, axonal, type 2P; CMT2G (a historically separate entity now **reclassified as CMT2P** after identification of the *LRSAM1* p.Cys694Tyr mutation — [PMID: 27686364](#)); *LRSAM1*-related Charcot-Marie-Tooth disease.

Information source type. The knowledge base for CMT2P is derived overwhelmingly from **aggregated disease-level resources** — OMIM, published family/pedigree studies, and case reports — rather than large EHR cohorts, reflecting its rarity.

2. Etiology

Primary cause — genetic. CMT2P is a monogenic Mendelian disorder caused by pathogenic variants in *LRSAM1*. The first description came from **Guernsey et al. (2010)** who used homozygosity mapping in a large recessive eastern Canadian kindred (locus chr9:122–129 Mb) and identified a homozygous intronic splice-acceptor variant (AG→AA) producing a frameshift/truncation and complete loss of protein ([PMID: 20865121](#)). **Weterman et al. (2012)** then identified a dominant frameshift, **p.Leu708Argfs*28**, in the C-terminal RING domain in a three-generation family with a strong LOD score of 5.12 ([PMID: 22012984](#)).

"A homozygous pathogenic variant was identified in the gene encoding leucine rich repeat and sterile alpha motif 1 (LRSAM1) by direct DNA sequencing" — Guernsey et al., [PMID: 20865121](#)

"This frameshift mutation (p.Leu708Argfx28) is located in the C-terminal RING finger motif of the encoded protein." — Weterman et al., [PMID: 22012984](#)

Genetic risk factors. The causal variants themselves are the risk factors; no additional susceptibility loci are established. **Modifier considerations:** in a family co-segregating a *RAB7A* (CMT2B) mutation and a novel *LRSAM1* variant, gender and possible *RAB7A/LRSAM1* gene interactions were proposed to explain marked variability in age of onset ([PMID: 27462242](#)).

Environmental risk / protective factors. No environmental risk or protective factors are established for CMT2P in humans. However, an important **gene–environment interaction** was demonstrated in the mouse model: *Lrsam1* mutant mice are hypersensitive to the axonal neurotoxin **acrylamide**, indicating that loss of LRSAM1 lowers the axonal threshold for degeneration under toxic stress ([PMID: 23519028](#)). This suggests that, mechanistically, neurotoxic exposures could plausibly aggravate an LRSAM1-compromised nervous system, though this has not been shown clinically.

Genetic protective factors. None specifically identified. Incomplete penetrance/asymptomatic carriers imply the existence of unknown modifiers, but none have been mapped.

3. Phenotypes

CMT2P phenotypes are those of a distal, length-dependent axonal sensorimotor polyneuropathy. Onset is typically **adult/late (2nd–5th decade)**, severity is **mild to moderate**, progression is **slow and progressive**, and there is **variable expressivity** including asymptomatic carriers ([PMID: 33568173](#), [PMID: 33414056](#)).

"dominant CMT2P is usually characterized by relatively mild, slowly progressive axonal neuropathy, mainly involving lower limbs, with age of onset between the second and fifth decades of life" — Palaima et al., [PMID: 33568173](#)

"CMT2P is a rare, but nevertheless relevant cause of adult-onset axonal and painful neuropathy" — Reilich et al., [PMID: 33414056](#)

Phenotype	Type	HPO term (suggested)	Onset / severity / progression	Notes
Distal lower-limb muscle weakness	Clinical sign	HP:0009053 / HP:0002460	Adult; mild-moderate; progressive	Legs affected before/more than arms
Distal sensory loss	Symptom/sign	HP:0002936 (distal sensory impairment)	Adult; variable	Length-dependent
Neuropathic / painful neuropathy	Symptom	HP:0009830 (peripheral neuropathy), HP:0012531 (pain)	Adult; variable	Notable feature of CMT2P
Pes cavus (high-arched foot)	Physical manifestation	HP:0001761	Insidious	Classic CMT deformity
Gait instability / disturbance	Clinical sign	HP:0002317 (unsteady gait), HP:0001288	Progressive	Major functional impact
Reduced/absent deep tendon reflexes	Clinical sign	HP:0001265 (areflexia) / HP:0001315	Early	Distal predominance
Foot drop	Clinical sign	HP:0009027	Adult	From distal weakness
Fatty atrophy of lower-limb muscle (MRI)	Laboratory/imaging	HP:0003693 (distal amyotrophy)	Subclinical detectable	Detects subclinical carriers (PMID: 27686364)
Parkinsonism (emerging)	Clinical sign	HP:0001300 (parkinsonism)	Late (5th–7th decade)	CNS extension (PMID: 26900582 , PMID: 40721190)

Quality-of-life impact. As with CMT broadly, the dominant burden is on **mobility and gait**, with foot deformity, falls risk, and — distinctively for CMT2P — **chronic neuropathic pain**. Supportive care (orthotics, rehabilitation) improves quality of life ([PMID: 40014417](#), [PMID: 40636623](#)). Formal EQ-5D/SF-36/PROMIS data specific to CMT2P are not available.

4. Genetic / Molecular Information

Causal gene. *LRSAM1* (Leucine Rich Repeat And Sterile Alpha Motif containing 1; also called **Tal**), HGNC:25135, gene OMIM 610933, NCBI Gene 90678, UniProt Q6UWE0, on **chromosome 9q33.3**. It encodes a universally expressed **RING-type E3 ubiquitin-protein ligase** with domain architecture **LRR – coiled-coil – SAM – PDZ-binding – RING** ([PMID: 33568173](#), [PMID: 23245322](#)).

Pathogenic variants cluster in the **3'-prime exons encoding the C-terminal RING domain**. Reilich et al. (2021) reported 14 patients from 12 families harboring 10 different *LRSAM1* variants (7 novel), predominantly dominant ([PMID: 33414056](#)).

Variant	Type	Inheritance	Source
Intronic splice-acceptor AG→AA (frameshift/truncation, complete LOF)	Splice-site	Recessive (homozygous)	PMID: 20865121
p.Leu708Argfs*28	Frameshift (RING domain)	Dominant	PMID: 22012984
p.Pro707Leu	Missense (RING)	Dominant	PMID: 28335037
p.Cys694Tyr	Missense (RING)	Dominant (reclassified CMT2G→CMT2P)	PMID: 27686364

Variant classification (ACMG/AMP). Interpretation is challenging precisely because pathogenic variants cluster at the 3' end where premature-termination codons **may or may not escape NMD**; careful application of PVS1 and related criteria is required depending on exact position and predicted NMD outcome ([PMID: 33414056](#)).

"Variants at the 3`end may or may not escape from nonsense-mediated decay, thereby defining the pattern of inheritance" — Reilich et al., [PMID: 33414056](#)

"clustering of pathogenic variants in 3'-prime exons, interpretation of genetic variants in LRSAM1 is challenging" — Reilich et al., [PMID: 33414056](#)

Functional consequences. - **Recessive** variants → complete loss of LRSAM1 protein → **loss of function.** - **Dominant** variants → truncated RING-domain protein escaping NMD → **dominant-negative** effect on ligase activity ([PMID: 33414056](#)).

"the C-terminal RING domain, which exerts a dominant-negative effect on protein function, whenever affected by an altered or truncated protein" — Reilich et al., [PMID: 33414056](#)

Allele frequency / somatic vs germline. All disease variants are **germline** and rare (not established as recurrent population polymorphisms); LRSAM1 mutations are considered rare even within CMT2 ([PMID: 28335037](#)). No somatic (COSMIC/cancer) role is relevant.

Modifier genes / epigenetics / chromosomal abnormalities. Possible *RAB7A* × *LRSAM1* interaction and a gender effect were proposed as phenotype modifiers ([PMID: 27462242](#)). No epigenetic mechanisms or large chromosomal abnormalities are implicated; transcriptomic changes (upregulation of *NEDD4L* and *TNFRSF21*) were seen with the p.Cys694Tyr mutation ([PMID: 27686364](#)).

5. Environmental Information

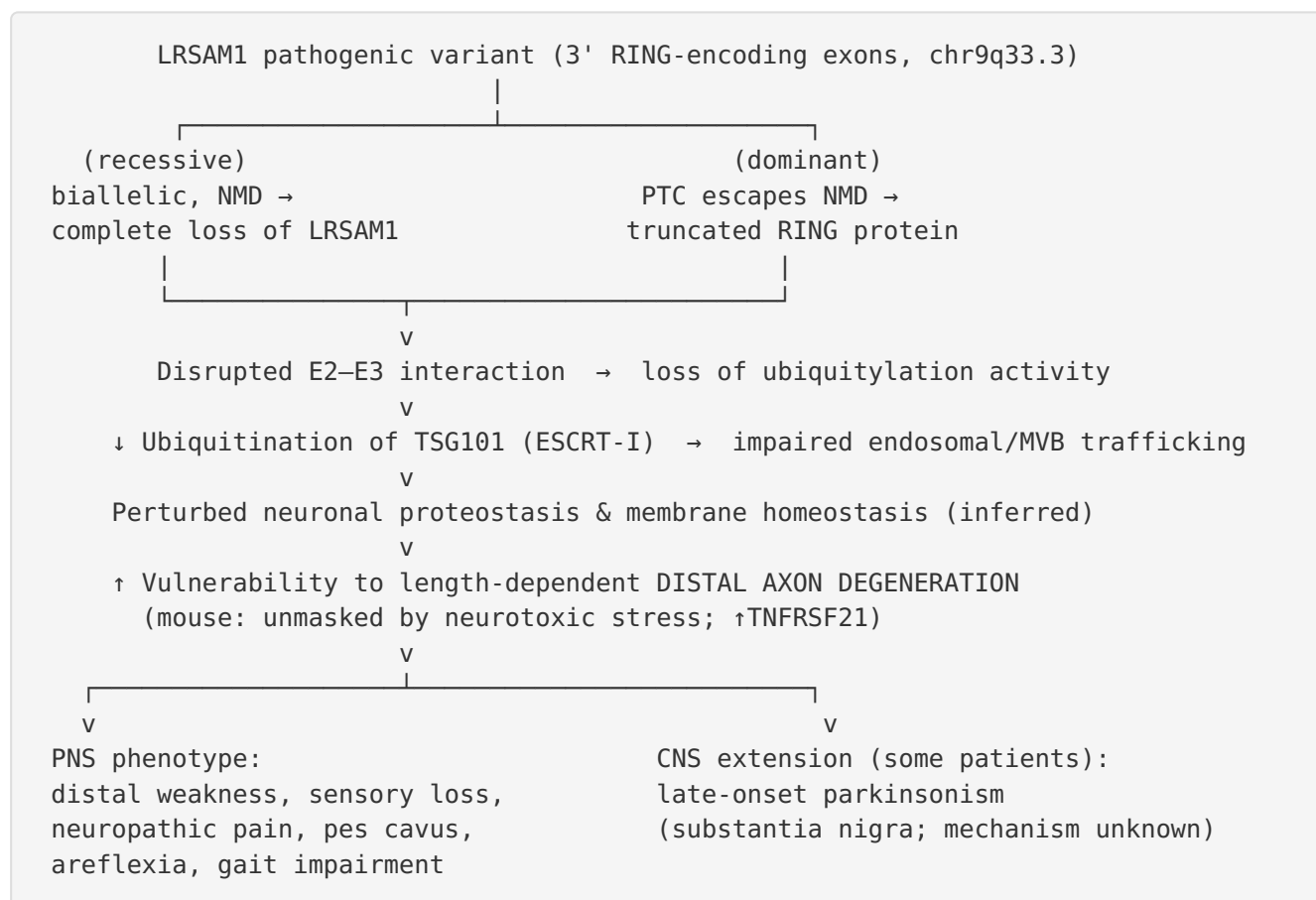
No environmental, lifestyle, or infectious agents cause CMT2P — it is a purely genetic disorder. The only mechanistically relevant environmental link is experimental: LRSAM1-deficient mouse axons are **more sensitive to the neurotoxin acrylamide** ([PMID: 23519028](#)), raising the theoretical possibility that neurotoxic exposures could exacerbate axonal vulnerability. Of interest to the protein's normal biology, LRSAM1 participates in **anti-bacterial autophagy (xenophagy)** of intracellular *Salmonella* Typhimurium ([PMID: 23245322](#)), but there is no evidence infection triggers the neuropathy.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A germline pathogenic variant arises in **LRSAM1**, clustered in the 3' exons encoding the **C-terminal RING domain**.
2. Depending on variant position and NMD, this **leads to** one of two states:
3. **(Recessive branch)** biallelic loss → NMD/complete absence of LRSAM1 protein → total **loss of E3 ligase function**; or
4. **(Dominant branch)** a truncated protein that escapes NMD → a **dominant-negative** RING-domain protein that poisons residual ligase activity.
5. Either state **results in** disrupted **E2–E3 interaction**, the essential step for transferring ubiquitin from the E2 conjugating enzyme to substrates — thereby **abrogating LRSAM1-mediated ubiquitylation** ([PMID: 28335037](#)).
6. Loss of ubiquitylation of the validated target **TSG101** (an ESCRT-I component) **impairs** endosomal sorting / multivesicular-body (MVB) trafficking and receptor endocytosis ([PMID: 15256501](#)) — *inferred* to perturb membrane/protein homeostasis in neurons.
7. Impaired proteostasis and trafficking **lowers the threshold** for **length-dependent distal axonal degeneration** — demonstrated indirectly by heightened axonal vulnerability to neurotoxic stress in the mouse model ([PMID: 23519028](#)).
8. Degeneration of the longest peripheral motor and sensory axons **produces** distal muscle weakness/atrophy, distal sensory loss, neuropathic pain, areflexia, pes cavus, and gait impairment — the CMT2P phenotype.
9. **(Branch — CNS extension, inferred)** In some individuals the same ligase dysfunction **is associated with** later degeneration affecting the substantia nigra, **manifesting as** late-onset parkinsonism ([PMID: 26900582](#), [PMID: 40721190](#)); the mechanism connecting LRSAM1 to nigral neurons remains undefined.

Causal chain diagram



Supporting detail

- **Molecular pathway / biochemical defect.** The core lesion is in the **ubiquitin–proteasome / ubiquitin-conjugation pathway**. LRSAM1 is a RING E3 ligase; RING-domain variants cause "*loss of the E2-E3 interaction that is an essential prerequisite for supporting ubiquitylation of target substrates*" (PMID: 28335037).
- **Protein function.** Two functional modules: the **LRR domain targets** substrates (e.g., cytosolic bacteria) and the **RING domain catalyzes** ubiquitin transfer — "*these functions require LRSAM1's leucine-rich repeat and RING domains, respectively*" (PMID: 23245322).
- **Cellular processes.** Endosomal/ESCRT trafficking, MVB and retrovirus budding (via TSG101 monoubiquitination, PMID: 15256501); autophagy/xenophagy; and, downstream, **axon degeneration**. Transcriptomic profiling of patient cells implicated *TNFRSF21* (a regulator of axonal degeneration) and pathways shared with ALS and Alzheimer disease (PMID: 27686364).

- **Subcellular localization.** In transfected cells LRSAM1 localizes to a **perinuclear compartment just beyond the Golgi** ([PMID: 23519028](#)).

Upstream vs downstream: the mutation and lost ubiquitylation are **upstream**; trafficking/ proteostasis defects are **intermediate**; distal axonal degeneration and clinical signs are **downstream**.

Suggested ontology terms. GO:0016567 (protein ubiquitination), GO:0004842 (ubiquitin-protein transferase activity), GO:0061630 (ubiquitin protein ligase activity), GO:0007032 (endosome organization), GO:0098930 (axonal transport), GO:0006914 (autophagy). **Cell types (CL):** CL:0000101 (sensory neuron), CL:0000100 (motor neuron), CL:0002573 (Schwann cell). **Cellular components:** GO:0005768 (endosome), GO:0000813 (ESCRT-I complex), GO:0005794 (Golgi apparatus).

7. Anatomical Structures Affected

Organ / system level. The **peripheral nervous system** is primary (UBERON:0000010). Body system: nervous system (UBERON:0001016). Secondary/musculoskeletal involvement includes distal limb muscles and the **foot/skeleton** (pes cavus, foot deformity); CMT patients also show an under-recognized rate of **hip acetabular dysplasia** (21% of imaged CMT patients in one series — [PMID: 40432997](#)). The emerging CNS extension implicates the **substantia nigra / basal ganglia** ([PMID: 26900582](#)).

Tissue / cell level. Nervous tissue — **peripheral motor and sensory axons**; LRSAM1 is abundantly expressed in PNS motor and sensory neurons ([PMID: 23519028](#)). Affected cells: **motor neurons (CL:0000100)** and **sensory neurons (CL:0000101)**; skeletal muscle secondarily denervated (CL:0000188).

Subcellular level. Endosome/MVB (GO:0005768), ESCRT-I complex (GO:0000813), Golgi-adjacent perinuclear compartment (GO:0005794).

Localization. Distal, length-dependent, **bilateral and symmetric**, lower limbs > upper limbs. Suggested UBERON: UBERON:0001322 (sciatic nerve) and distal peroneal/tibial nerve territory; spinal cord and CNS not primarily affected in the classic peripheral phenotype.

8. Temporal Development

- **Onset:** adult/late, typically **2nd to 5th decade**; **insidious/chronic** (PMID: 33568173). A rare **early-onset autosomal dominant** family has been reported (PMID: 38330802).
- **Progression:** **slow and progressive**; generally mild ("quiescent") axonal neuropathy (PMID: 27686364).
- **Course pattern:** chronic, lifelong, non-remitting, non-episodic.
- **Critical periods / CNS window:** late-onset parkinsonism appears **years after** neuropathy onset (ages ~50–65) in some patients (PMID: 26900582).

9. Inheritance and Population

Epidemiology. CMT overall affects ~**1:2,500** worldwide with >100 known genetic causes (PMID: 40219666). CMT2P is a **rare** subtype; *LRSAM1* mutations are uncommon even within CMT2 (PMID: 28335037). Precise CMT2P prevalence/incidence are not established.

"Affecting about ~1:2,500 people worldwide, CMT has over 100 known genetic causes, leading to different subtypes with varying disease severity and progression." — Stavrou et al., PMID: 40219666

Inheritance. Both **autosomal dominant (predominant)** and **autosomal recessive**, determined by variant location and NMD outcome (see Section 4). Dominant = dominant-negative RING variants escaping NMD; recessive = biallelic complete loss (PMID: 33414056).

Penetrance / expressivity. **Incomplete penetrance** and **variable expressivity**, including asymptomatic carriers detectable only by lower-limb muscle MRI (PMID: 27686364, PMID: 33568173).

Founder effects / consanguinity. The original recessive kindred was a large **consanguineous eastern Canadian** family (homozygous founder-type splice variant) (PMID: 20865121). Carrier status for the recessive *LRSAM1* allele has been incidentally detected in idiopathic neuropathy cohorts (PMID: 39290488).

Demographics. No strong ethnic predilection; reported families span Canada, the Netherlands, and others. Sex ratio not clearly skewed, though gender was proposed as a modifier in one family ([PMID: 27462242](#)). Genetic anticipation and germline mosaicism are not established.

10. Diagnostics

Clinical / electrophysiology. Diagnosis rests on **nerve conduction studies / EMG** demonstrating an **axonal** (rather than demyelinating) sensorimotor neuropathy: reduced amplitudes with relatively preserved conduction velocities ([PMID: 40721190](#)). Clinical exam shows distal weakness, sensory loss, areflexia, pes cavus.

Imaging. **MRI of lower-limb musculature** systematically reveals **fatty atrophy** in both clinical and subclinical carriers and is useful to detect minimal/subclinical disease ([PMID: 27686364](#)).

Genetic testing (definitive). Diagnosis is confirmed by identifying a pathogenic *LRSAM1* variant. Recommended approach: **multigene neuropathy/CMT panel or exome sequencing**, given the >100 CMT genes; targeted *LRSAM1* analysis when phenotype/family history suggests it. *LRSAM1* is included on inherited neuromuscular gene panels ([PMID: 39290488](#)). Careful ACMG interpretation of 3' PTC variants (NMD escape) is essential ([PMID: 33414056](#)). Chromosomal microarray, karyotyping, FISH, mitochondrial and repeat-expansion testing are **not** relevant to CMT2P.

Omics diagnostics. Not standard; research transcriptome profiling of patient lymphoblasts has shown disease-associated expression changes ([PMID: 27686364](#)).

Differential diagnosis. Other axonal CMT2 subtypes (CMT2A/*MFN2*, CMT2B/*RAB7A*, CMT2K/*GDAP1*, etc.), hereditary transthyretin (ATTRv) amyloidosis (important treatable mimic — [PMID: 39290488](#)), CMT1 (demyelinating; *PMP22*), acquired/idiopathic axonal polyneuropathies, and ALS5/SPG11 overlap syndromes ([PMID: 26556829](#)).

Screening. Cascade genetic testing of at-risk relatives once a familial variant is known; no population newborn screening.

11. Outcome / Prognosis

- **Survival / life expectancy: normal.** CMT2P is not life-limiting; by analogy to other CMT the disease "*reflects a process of normal ageing*" with **normal life expectancy** (PMID: [24646194](#)).
 - **Morbidity / disability.** Chronic, slowly progressive **mobility impairment**, foot deformity, falls risk, and **neuropathic pain** dominate; wheelchair dependence is uncommon given the mild course.
 - **Complications.** Foot deformities requiring surgery; possible hip acetabular dysplasia in CMT broadly (PMID: [40432997](#)); **late-onset parkinsonism** as a distinct CNS complication in some LRSAM1 patients (PMID: [26900582](#)).
 - **Prognostic factors.** Variant type/inheritance (dominant-negative vs LOF), and modifiers such as gender/second-gene interaction (PMID: [27462242](#)). No validated molecular prognostic biomarkers.
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12. Treatment

No curative therapy exists; management is symptomatic and supportive.

"Currently, no curative treatment exists for CMT. Management focuses on symptomatic interventions, including orthotic support, surgical procedures, and physical therapy." — Kikuchi, PMID: [40636623](#)

Modality	Intervention	Suggested NCIT
Orthotics	Ankle-foot orthoses (AFOs) for foot drop	NCIT:C50008 (orthotic device)
Rehabilitation	Physical therapy, occupational therapy	NCIT:C15327 (physical therapy)
Surgical	Correction of pes cavus / foot deformity	NCIT:C15329 (surgery)
Pharmacologic (supportive)	Neuropathic pain control (gabapentinoids, duloxetine, TCAs)	NCIT:C15224 (pain management)
Genetic counseling	Family risk assessment	NCIT:C15633 (genetic counseling)

Experimental / disease-modifying (CMT broadly, not LRSAM1-specific). No approved disease-modifying drug for CMT2P. Across CMT, **gene therapy** (silencing/replacement/editing via AAV vectors) is under active preclinical/early-clinical development but faces blood–nerve/brain-barrier delivery, immunogenicity, and scalability challenges ([PMID: 40219666](#)). Other emerging approaches include **PMP22 silencers** (for CMT1A), **HDAC6 inhibitors**, and **govorestat** (aldose reductase inhibitor, CMT-SORD trial) — none specific to LRSAM1 ([PMID: 40014417](#)).

"Currently, there are no approved treatments and care focuses on managing symptoms."
— Stavrou et al., [PMID: 40219666](#)

Pharmacogenomics / personalized medicine. No LRSAM1-specific pharmacogenomic guidance. The dual LOF-vs-dominant-negative mechanism implies **different rational strategies** (gene replacement for recessive LOF; allele-specific silencing for dominant-negative alleles) — currently conceptual.

13. Prevention

CMT2P cannot be prevented (genetic). Prevention is limited to **reproductive/genetic risk management** and **complication avoidance**:

- **Primary:** Genetic counseling, carrier/cascade testing, and reproductive options (prenatal testing, preimplantation genetic diagnosis) for known familial variants (general CMT precedent — [PMID: 24646194](#)).
- **Secondary:** Early detection of subclinical carriers via lower-limb muscle MRI ([PMID: 27686364](#)); early physiotherapy/orthotics.
- **Tertiary:** Prevent complications — orthotic support to prevent falls/deformity progression, monitoring for hip dysplasia and, given the emerging link, parkinsonian features.
- **Behavioral:** Theoretical avoidance of neurotoxic exposures (extrapolated from acrylamide sensitivity in the mouse model, [PMID: 23519028](#)) — not a clinical guideline.

No immunization, public-health, or infectious-control measures apply.

14. Other Species / Natural Disease

- **Model species:** *Mus musculus* (NCBI Taxon 10090) — the *Lrsam1* mutant mouse ([PMID: 23519028](#)).
 - **Orthologous gene:** mouse *Lrsam1*, abundantly expressed in PNS motor and sensory neurons.
 - **Natural disease in companion animals/wildlife:** No naturally occurring LRSAM1/CMT2P disease documented in other species (OMIA); the LRR–RING E3-ligase function is **evolutionarily conserved**, and the protein's ubiquitin-dependent xenophagy role is conserved across mammalian cells ([PMID: 23245322](#)).
 - **Zoonotic potential:** None (non-infectious genetic disease).
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15. Model Organisms

Mouse (*Lrsam1* mutant). The principal model ([PMID: 23519028](#)):

- **Phenotype recapitulation (partial):** Both homozygous and heterozygous mice have **largely normal neuromuscular performance and only very mild age-related neuropathy** — milder than the human disease.
- **Key finding — sensitized axons:** mutant axons are hypersensitive to axonal neurotoxin challenge.

"Lrsam1 mutant mice are more sensitive to challenge with acrylamide, a neurotoxic agent that causes axon degeneration, indicating that the axons in the mutant mice are indeed compromised" — Bogdanik et al., [PMID: 23519028](#)

- **Localization insight:** LRSAM1 localizes to a **perinuclear post-Golgi compartment**, with little colocalization with endosome-to-lysosome trafficking machinery.
 - **Limitations:** Mild phenotype limits modeling of overt human neuropathy; does not capture dominant-negative human biology or the parkinsonism extension.
 - **Cellular / in vitro models:** Patient lymphoblasts/fibroblasts for ubiquitylation assays and transcriptomics ([PMID: 28335037](#), [PMID: 27686364](#)); in vitro ubiquitylation reconstitution to test E2–E3 interaction defects.
 - **Resources:** MGI (mouse) and standard CMT model repositories.
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Mechanistic Model / Interpretation

The unifying theme is **failure of a RING E3 ubiquitin ligase**, converging on impaired proteostasis and endosomal trafficking in neurons. CMT2P is mechanistically distinctive within CMT2 for two reasons: (a) its **position-dependent dual inheritance**, in which NMD escape/triggering of a 3' premature-termination codon decides whether disease is dominant (dominant-negative) or recessive (complete loss of function); and (b) a **dominant-negative** mode operating for most disease alleles. The recurrent link to parkinsonism situates LRSAM1 within broader **ubiquitin-proteostasis neurodegeneration** biology — echoed by patient-cell

transcriptomes sharing pathways with ALS and Alzheimer disease — and suggests the same ligase defect can, over decades, extend from the longest peripheral axons to vulnerable central (nigral) neurons.

Evidence Base

PMID	Study	Contribution
20865121	Guernsey et al. 2010	First identified <i>LRSAM1</i> as causal (recessive splice variant, complete LOF)
22012984	Weterman et al. 2012	Dominant RING frameshift p.Leu708Argfs*28 (LOD 5.12)
28335037	Hakonen et al. 2017	Biochemical defect: RING variants disrupt E2–E3 interaction, abrogate ubiquitylation
33414056	Reilich et al. 2021	Genotype–phenotype: 3' clustering, NMD determines inheritance, dominant-negative RING
33568173	Palaima et al. 2021 (review)	Clinical spectrum; TSG101 as main substrate; domain architecture
27686364	2016	CMT2G reclassified as CMT2P (p.Cys694Tyr); MRI subclinical detection; transcriptome
23519028	Bogdanik et al. 2013	Mouse model: mild phenotype, acrylamide sensitization, localization
15256501	Amit et al. 2004	<i>LRSAM1</i> /Tal ubiquitinates TSG101; ESCRT/MVB & retrovirus budding
23245322	Huett et al. 2012	LRR (targeting) + RING (catalytic) domains; anti-bacterial xenophagy
26900582	Aerts et al. 2016	Parkinsonism in 3/5 affected family members — CNS extension
40721190	Ducatel et al. 2025	CMT2P with parkinsonism case — corroborates CNS link
40219666	Stavrou et al. 2025	CMT prevalence (1:2,500); no approved treatments; gene-therapy status
40636623	Kikuchi 2025	Symptomatic management (orthotics, PT, surgery)
40014417	De Grado et al. 2025	Current CMT therapeutics landscape / trials
38330802	2024	Early-onset autosomal dominant CMT2P family
27462242	2016	Gender & <i>RAB7A</i> / <i>LRSAM1</i> interaction as phenotype modifiers
39290488	2024	<i>LRSAM1</i> carrier detected on neuropathy gene panel

PMID	Study	Contribution
24646194	2014	CMT precedent: normal life expectancy, symptomatic care
40432997	2025	Hip acetabular dysplasia in CMT (21%)
26556829	2015	ALS5/SPG11 overlap — differential diagnosis context

Evidence source types: human clinical/genetic (family & cohort studies, reviews), model organism (mouse), and in vitro/biochemical (ubiquitylation assays, cell transfection).

Limitations and Knowledge Gaps

1. **Rarity limits epidemiology.** No reliable prevalence/incidence, sex ratio, or natural-history quantitation specific to CMT2P; most data come from individual pedigrees and case reports.
2. **Substrate biology is incomplete.** TSG101 is the only well-validated LRSAM1 ubiquitylation target relevant to disease; how impaired TSG101/ESCRT function specifically drives *axonal* degeneration is inferred, not demonstrated.
3. **Genotype–phenotype uncertainty.** NMD escape/dominant-negative predictions for 3' PTC variants are not always experimentally verified, complicating ACMG classification.
4. **Mouse model is mild.** The *Lrsam1* mouse under-recapitulates overt human neuropathy and does not model the dominant-negative or parkinsonism aspects.
5. **The parkinsonism link is preliminary.** Based on one family plus isolated cases; causality, penetrance, and mechanism (LRSAM1 in substantia nigra) are unknown.
6. **No disease-specific therapy or biomarkers.** All treatments are symptomatic; no validated prognostic molecular markers.

Proposed Follow-up Experiments / Actions

1. **Functional NMD/dominant-negative assays** for each reported 3' *LRSAM1* PTC variant (patient RNA + minigene assays) to firm up ACMG calls and inheritance prediction.

2. **iPSC-derived motor/sensory neurons** carrying dominant-negative vs LOF *LRSAM1* alleles to test axonal degeneration, ESCRT/TSG101 trafficking, and stress vulnerability — bridging the mild mouse-model gap.
3. **Expand substrate mapping** (ubiquitylation proteomics) in neurons to identify axon-relevant LRSAM1 targets beyond TSG101.
4. **Systematic ascertainment of parkinsonism/CNS features** across LRSAM1 cohorts (DAT-SPECT imaging, longitudinal follow-up) to establish frequency, penetrance, and mechanism of the CNS extension.
5. **Allele-specific silencing (dominant alleles) and gene replacement (recessive LOF)** proof-of-concept in cellular/mouse models, leveraging the AAV gene-therapy platforms being developed for CMT broadly.
6. **Registry-based natural history study** to define CMT2P prevalence, progression rate, and disability trajectory, enabling future trial readiness.
7. **Test neurotoxic-exposure interaction** clinically/epidemiologically, motivated by acrylamide hypersensitivity in the mouse model.

Report compiled from an autonomous multi-iteration literature investigation (8 confirmed findings, 21 papers reviewed). Evidence is predominantly human genetic/clinical and model-organism; CMT2P remains a rare, symptomatically managed axonal neuropathy with an emerging CNS dimension.
